

Wandering Black Hole Feeds on Its Own Gravitational Wake

A 35,000-solar-mass intermediate-mass black hole accretes from a dense downstream wake, revealing a new growth channel.

arXiv:2608.10719 — Multi-wavelength evidence for off-nuclear IMBH accretion via gravitational focusing.

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[arXiv:2608.10719](#) [doi:10.48550/arXiv.2608.10719](#)



Background

Intermediate-mass black holes are seeds of supermassive black holes, but many wander off-nuclear after galaxy mergers. Their fuel supply is uncertain because standard feeding channels are inaccessible.

Direct observational evidence for wandering black holes accreting from gravitational wakes has remained elusive.

What this paper is selling

This paper reports a 35,000-solar-mass wandering black hole accreting through a gravitational wake. Multi-epoch spectroscopy reveals upstream flow, downstream wake, and optically thick absorbers driving changing-look variability.

Key results to remember

- Broad lines, compact continuum, variability, and power-law SED confirm black hole nature.
- Three gas components: blueshifted upstream flow, redshifted dense downstream wake, optically thick absorbers.
- Absorbers within capture radius drive rapidly changing-look broad-line variability.

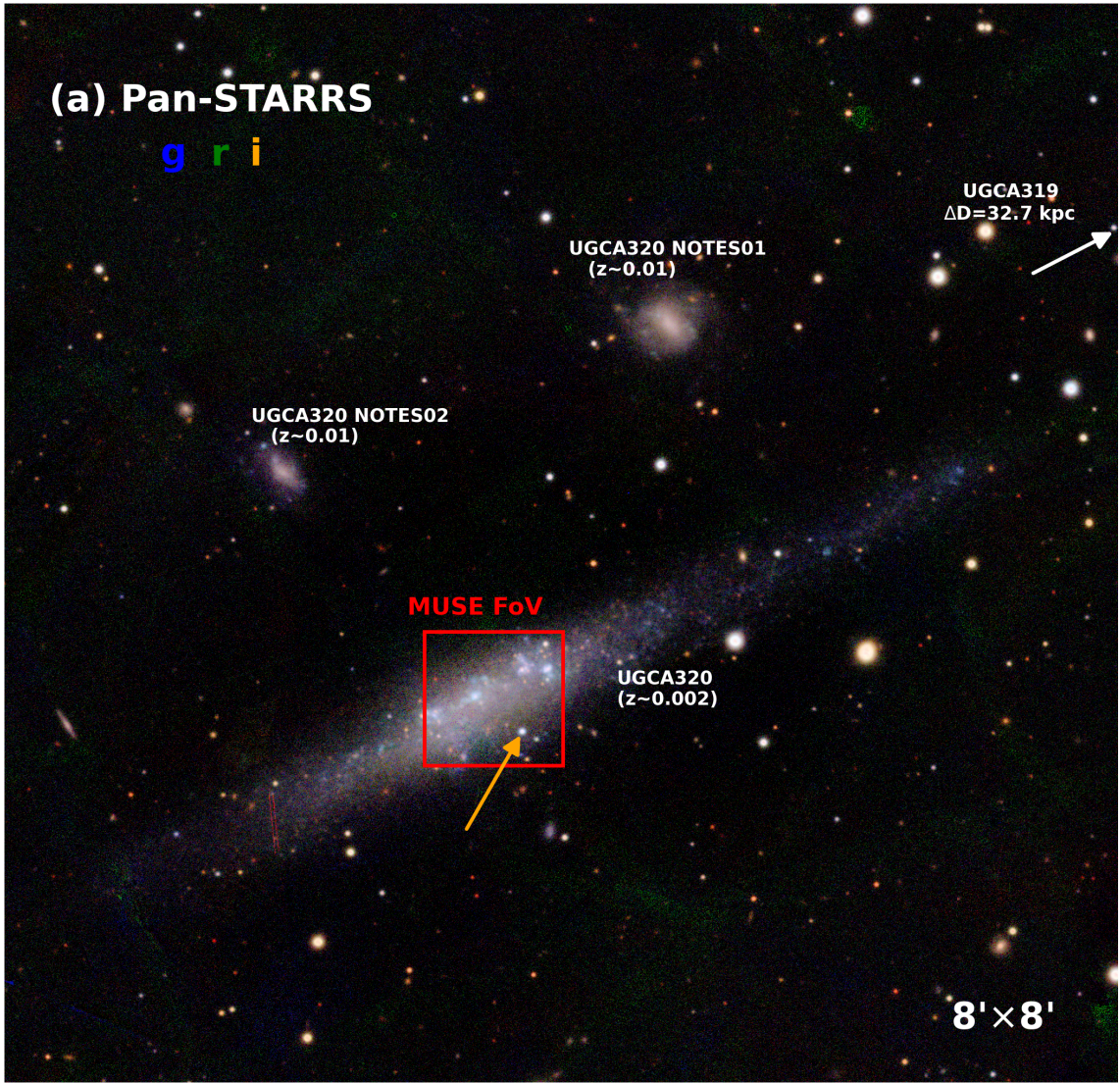


Fig. 1. HST and MUSE views pinpoint the off-nuclear IMBH in UGCA 320, with...

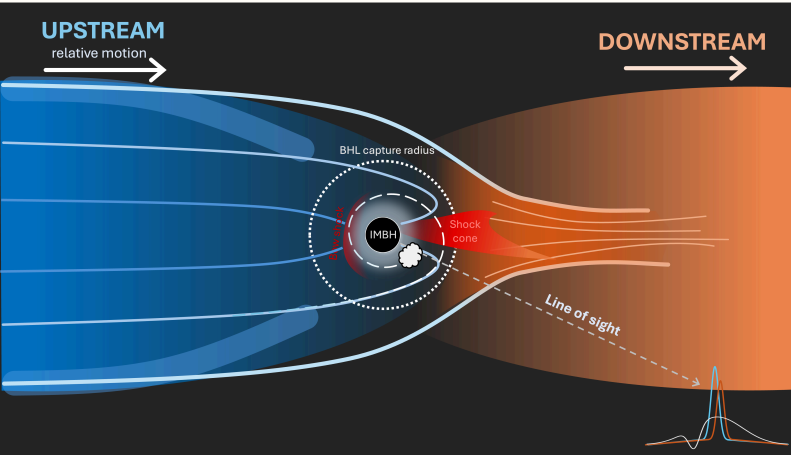


Fig. 4. Schematic of Bondi–Hoyle–Lyttleton accretion: upstream gas focuses into a dense wake, feeding...